

Michael Do

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michaeldo2004.github.io/PersonalSite/

EDUCATION

California State University, Fullerton
Bachelor of Science in Computer Science
GPA: 3.65

08/2022 – 05/2026 | Fullerton, CA

EXPERIENCE

Assure-US Research, Data Research Assistant

06/2023 – 07/2023 | Fullerton, CA

- Analyzed SAT, ACT, and GPA data from 9,000+ students using **pandas** and **NumPy** to identify relationships between standardized test performance and academic outcomes.
- Built and evaluated **linear regression models** with **scikit-learn** to predict student GPA.
- Presented insights and model results through structured Jupyter Notebook presentations to a large academic audience.

PROJECTS

Connect4Web | React, Flask, Socket.IO, Supabase, Python

07/2026

- Built a **full-stack** Connect 4 web application supporting authenticated single player and real-time multiplayer gameplay, reconnect handling, game persistence, and post-game analysis.
- Implemented and evaluated an **iterative-deepening Minimax** algorithm with **alpha-beta pruning**, heuristic board evaluation, time-bounded search, and a 12,000-state **transposition cache** across four difficulty levels.
- Leveraged Claude and OpenAI Codex for **architecture analysis, implementation, debugging, and test generation**, verifying AI-assisted changes through 187 automated tests and shared JSON Schema contracts.

Logic Grid Puzzle (Capstone Project) | TypeScript, Python, SQL

05/2026

- Designed and implemented a full-stack **leaderboard system** using **SvelteKit** and **Django REST Framework** with filtering, sorting, and paginated results, displaying contextual rank and top-percentile stats for authenticated users
- Built a user profile dashboard in **SvelteKit** aggregating personalized solve history with searchable, filterable statistics, leveraging Svelte 5 reactive state management
- Architected a **relational database schema** in PostgreSQL via **Supabase**, designing normalized tables with foreign key constraints across 6+ tracked fields per record including puzzle metadata and solve time
- Developed end-to-end **user authentication** with **Google OAuth 2.0**, email/password login, and session management using **Django REST Framework**, with client-side auth state handled via Svelte stores and input sanitization

DeepDelta (FullyHacks 2026) | Next.js, React, TypeScript, Tailwind CSS

04/2026

- Architected a Retrieval-Augmented Generation (RAG) pipeline using **Next.js, TypeScript, Anthropic API** to ground LLM responses against a curated knowledge corpus, reducing hallucination
- Engineered a **structured output** validation system using **Claude** with forced **tool-use**, producing a weighted 0–100 confidence score surfacing gaps, conflicts, and decay flags
- Built a full-stack comparison UI contrasting raw Claude, GPT-4o, and Gemini 2.5 Flash outputs against verified, source-cited responses via **REST API integration**
- Ensured fault tolerance by integrating **Human Delta** vector and file system search APIs with graceful **fallback mechanisms** across retrieval backend failures

Collaborative Whiteboard | React, FastAPI

11/2025

- Built real-time collaborative whiteboard with chat function using **FastAPI and React (Vite)**, extending Excalidraw functionality.
- Secured the application with **JWT-based authentication via Supabase**, including **role-based access authorization**.
- Designed and implemented **RESTful FastAPI endpoints** for projects, whiteboards, chat history, and user onboarding.
- Implemented **CRUD operations and persistent state** using Supabase database functions and RPCs, enabling collaborator invitations and multi-user sessions.

CERTIFICATES

Google Cloud Computing Foundations Certificate 

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional 

TECHNICAL SKILLS

Programming Languages: C++, Python, Java, MySQL, Assembly, TypeScript, JavaScript, HTML/CSS

Platforms: Linux, Windows, Jupyter Notebook, Google Cloud

Tools and Libraries: Visual Studio Code, Git, Flask, Numpy, React, Vite, FastAPI, Supabase, Next.js, Tailwind CSS, Anthropic SDK, OpenAI API, Gemini API, Django REST Framework, SvelteKit